

Final Project: Association between Student Behaviors and Exam Score

Itzel Aleman Flores and Kimberly By Goytia

Abstract

Previous research conducted by the NIH and CDC has found that several student habits can affect their academic performance, such as mental health, diet, and physical activity. This study will further explore what additional behaviors significantly affect a student's exam score, then it will delve into the complicated relationships that exist between two behaviors such as time spent studying and a student's mental health. This analysis was done via nested F-tests by running ANOVA analyses on linear and quadratic models. The tests revealed that among our statistically significant predictors, the hours spent studying had a quadratic relationship with predicted exam score. It was also found that the time spent studying and a student's mental health rating have a quadratic interaction relationship that influences predicted exam scores. Overall, our analysis found several interesting relationships between student habits that have statistically significant impacts on academic performance.

Introduction

Current reports show that some student behaviors influence academic outcomes, including physical activity, mental health, diet, sleep, and many more. We can see the direct benefits of positive health behaviors with academic success when it comes to healthy student diets via a CDC study, which determined that, "66% of U.S. high school students with mostly A's ate vegetables compared to 52% of students with mostly D's/F's," (Centers for Disease Control and Prevention 2019). However, not all factors that influence academic outcomes are as straightforward in their associations. Most health professionals encourage students to get high-quality sleep to improve their academic performance, but a study by the NIH saw a decline in GPA as hours of sleep increased beyond 9 hours. More specifically, the NIH discovered, "Students who slept 4 [hours] or less per night had lower GPAs than students who slept [6, 7, and 8 hours] per night. Students who slept 9 [hours] per night had lower GPAs than students who slept 8 [hours] per night," (Reuter and Forster 2021). Although sleep quality is an essential biological factor that plays a crucial role in educational achievement, obtaining too

much of it can cause a deterioration in a student's GPA. In addition to non-linear factors, like sleep, multiple health behaviors can impact one another and change the performance outcomes of a student. For example, it is well documented that low physical activity and low mental health both separately lead to a decrease in educational achievement. However, when students began to engage in increased physical activity, researchers saw improvement in their mental health, specifically noting "mood, attention, calmness, and a release of excess energy," which correlated with increased academic performance (Shaw et al. 2015).

Based on this current research landscape we hope to uncover what additional factors have a statistically significant impact on a student's performance, does too much of a good behavior (such as study hours) begin to hinder or plateau their exam score, and are there behaviors that interplay with each other influencing changes in exam performance. To uncover these nuances, we will focus on fitting a quadratic regression model on key statistically significant student behaviors and include interactions between factors to determine the significance of interplay. For this study, academic performance will be determined by a student's exam score on a scale from 0-100 instead of GPA or other measures of academic success. From our results, we hope to offer a nuanced understanding of what singular and interplaying student behaviors educators should target to create a well-balanced educational environment that sets students up for success.

Methods: Data and Analysis Variables

The target for our statistical analysis is students within educational systems that are similar to the one included in the dataset. It should be noted that the sampled population consists of simulated students based in the American educational system. The data we will be utilizing as the basis of our research comes from Kaggle, titled Student Habits vs Academic Performance created by Jayanta Nath (Nath 2025). It includes a variety of lifestyle habits that affect a student's overall academic performance, measured by their resulting exam score. We will be utilizing exam scores (0-100) as a function of all the variables within the data set to fit a model to determine which variables are significantly associated with predicting any given student's exam score. The variables are included in the table below with their corresponding meanings and units of measurement/ levels for the variables that are categorical.

Variable	Description
study_hours_per_day	average daily study time in hours
social_media_hours	daily social media time in hours
netflix_hours	average daily Netflix/ binging time in hours
part_time_job	whether the student has a part time job or not
attendance_percentage	classes the student attend represented as a percentage
sleep_hours	average daily sleep in hours
diet_quality	indicator of a poor, fair, or good diet

exercise_frequency	how times a week the student exercises
parental_education_level	highest form of education the student's parents have
internet_quality	indicator of a poor, average, or good internet quality
mental_health_rating	mental health rating of the student measured on a scale of 1 to 10
extracurricular_participation	whether the student participates in extracurriculars or not
exam_score	the final exam score on a scale of 0-100

Full list of variables included in the data set

Since the data set was simulated, there was no missing data, so it was not altered in a significant way. Additionally, we decided to proceed with an untransformed model to gain an understanding of our research questions, as there were no transformations that made a significant improvement on linearity and equal variance. Therefore, we determined that the violations were not severe enough to disregard the data.

Methods: Statistical Methodology

To find the most statistically supported model that helps us identify which student behaviors predict low exam scores, we fit a multiple linear regression model and conducted a nested F-test. We first fit a full model with all the possible predictors (such as daily hours of sleep, diet quality, mental health rating, etc.). Then, we used backward elimination at the 0.05 significance level to identify which variables were statistically significant and found our reduced model based on those variables. An ANOVA analysis was then run to compare the two models, in which the reduced model proved to be statistically supported for use in our analysis. The population and mathematical model with the statistically significant predictors are shown below:

Reduced Population Model: $E[\text{examscore}] = \beta_0 + \beta_1(\text{sleep}) + \beta_2(\text{socialmedia}) + \beta_3(\text{study}) + \beta_4(\text{netflix}) + \beta_5(\text{exercise}) + \beta_6(\text{mentalhealth}) + \beta_7(\text{attendance})$

Reduced Mathematical Model: $E[\text{examscore}] = 6.15722 + 2.00462(\text{sleep}) - 2.61978(\text{socialmedia}) + 9.57456(\text{study}) - 2.27708(\text{netflix}) + 1.45187(\text{exercise}) + 1.94891(\text{mentalhealth}) + 0.14473(\text{attendance})$

From our statistically significant predictors, we chose to delve into the relationship between study hours and exam score. Through visual analysis, we noticed that there seemed to be a quadratic relation between these two variables. To determine if this relationship was truly occurring, we compared a linear regression model and a quadratic model with study hours as the only predictor of exam score. By evaluating the p-values between each model, we determined that there was statistically significant evidence to support that study hours and exam scores have a quadratic relationship. The exact p-value evaluation will be expanded upon later in our paper.

Quadratic Population Model: $E[\text{examscore}] = \beta_0 + \beta_1(\text{studyhours}) + \beta_2(\text{studyhours}^2)$

Quadratic Mathematical Model: $E[examscore] = 33.7693 + 10.9228(studyhours) - 0.1995(studyhours^2)$

Now that we know the association between study hours and exam score is quadratic, we were curious to understand if other variables were interrelated with study hours and had an influence on exam score. From the original full model that contained only statistically significant predictors, we decided to look into the potential interactive relationship between study hours and mental health. This choice was informed by Shaw et al. (2015), whose research showed that mental health influences a student's resulting academic performance. Additionally, they go on to state that when a student increases their physical activity, it has a positive association with mental health, which relates to a positive change in academic performance. Due to this foundational research and our findings that mental health is influential to exam scores, we wanted to see if there existed a relation between mental health and study hours similar to the aforementioned interaction with physical activity and mental health. To test this association, we created a non-interactive and an interactive quadratic model, which both contained mental health as an added predictor.

After running an ANOVA F-test comparing both models, we found that our interactive model was statistically significant. However, on a p-value level within the quadratic interaction, we noticed a few discrepancies, which we will expand on in the following section.

Interactive Population Model: $E[examscore] = \beta_0 + \beta_1(studyhours) + \beta_2(studyhours^2) + \beta_3(mentalhealth) + \beta_4(mentalhealth * studyhours) + \beta_5(mentalhealth * studyhours^2)$

Interactive Mathematical Model: $E[examscore] = 24.10556 + 9.36587(studyhours) + 0.12280(studyhours^2) + 1.75199(mentalhealth) + 0.29671(mentalhealth * studyhours) - 0.06002(mentalhealth * studyhours^2)$

Results

When conducting data exploration, we noted that our data set consisted of 1,000 simulated students, and, as mentioned previously, no observations were removed or altered. First, we wanted to understand what predictors had a significant influence on a student's predicted exam score. To answer this question, we compared a full model and a reduced model. The reduced model was the most statistically supported model to proceed with compared to the full model ($p = 0.7382$, $f\text{-stat} = 0.7013$) as its AIC and BIC were smaller when compared to the full model. Both models had significant adjusted R^2 values (0.90). Therefore, the reduced model was more parsimonious and aligned with our research question.

Coefficients	Estimates	Standard Error	t-value	p-value
(Intercept)	6.15722	1.89252	3.25345	0.00118
sleep_hours	2.00462	0.13764	14.56388	0.00000
social_media_hours	-2.61978	0.14413	-18.17667	0.00000
study_hours_per_day	9.57456	0.11503	83.23819	0.00000
netflix_hours	-2.27708	0.15697	-14.50664	0.00000
exercise_frequency	1.45187	0.08338	17.41290	0.00000
mental_health_rating	1.94891	0.05924	32.89691	0.00000
attendance_percentage	0.14473	0.01797	8.05427	0.00000

Displays summary statistics of the reduced model

Based on our final reduced model, we can conclude that a one-hour average increase in daily social media usage is associated with an average 2.62-point decrease in exam score. A one-hour increase in daily Netflix usage is associated with an average 2.28-point decrease in exam score. Daily sleep, study hours, exercise, mental health rate, and lecture attendance are all associated with increasing exam scores. Specifically, a one-hour increase in daily sleep is associated with a 2.00 point increase in exam score. An increase in attending one lecture is associated with a 0.14 point increase in exam score. An increase of one exercise session a week is associated with a 1.45 point increase in exam score. A one-point increase in mental health rating is associated with a 1.95-point increase in exam score. Lastly, a one-hour increase in study hours is associated with a 9.57-point increase in exam score. From these associations, we wanted to see if too much of a good behavior could cause a decrease or plateau in exam score benefits. After plotting multiple predictors with exam score, we found that there seemed to be a plateau in exam score results and study hours seen in the following graph.

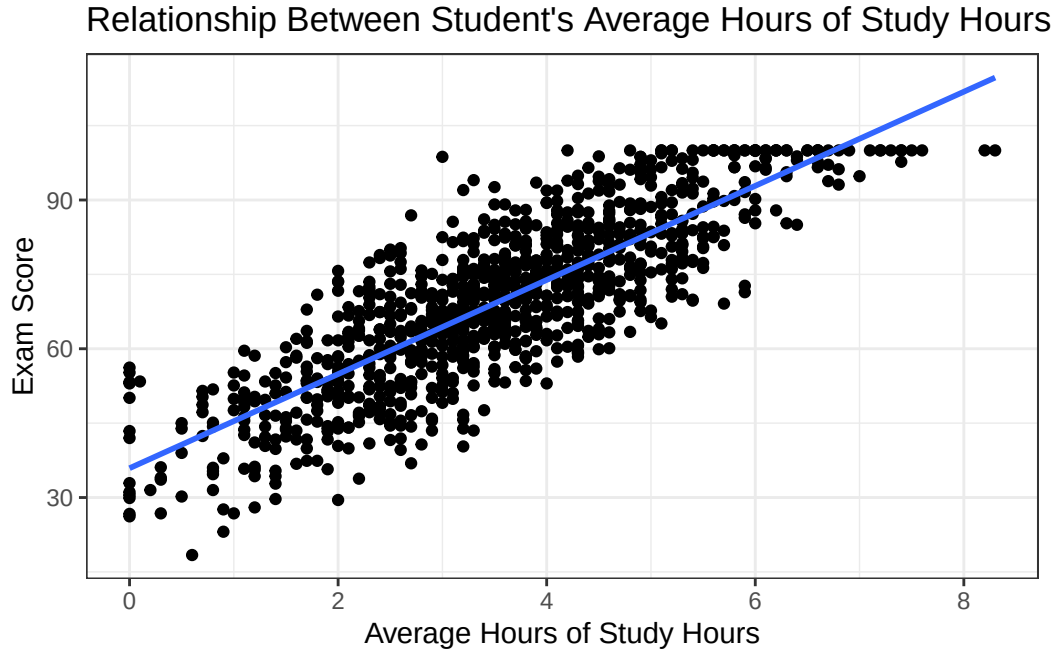


Figure 1: Displays relationship between study hours and exam score

The observed plateau between hours studied and exam score suggested that if a student studied for too many hours, they would see no additional increase in exam scores. This association indicated a potential non-linear relationship between study hours and exam scores.

When we assessed the statistical significance between our linear and quadratic models relating study hours to exam score, we noted that the quadratic study term was significant with a p-value of 0.0472 compared to an alpha of 0.05. This result led us to believe that there is a statistically significant quadratic relationship between the number of hours a student studies and their associated exam score.

Table 2: Table: Linear Regression Model for Study Hours and Exam Score

Coefficients	Estimates	Standard Error	t-value	p-value
(Intercept)	35.91016	0.78930	45.49618	0
study_hours_per_day	9.49025	0.20546	46.19119	0

Table 3: Table: Quadratic Model for Study Hours and Exam Score

Coefficients	Estimates	Standard Error	t-value	p-value
(Intercept)	33.76928	1.33513	25.29297	0.00000
study_hours_per_day	10.92276	0.74972	14.56920	0.00000
I(study_hours_per_day^2)	-0.19952	0.10044	-1.98656	0.04724

In particular, as a student's hours of study increased past a certain point, there were diminishing returns or a 0.1995 point decrease in the student's resulting score. We can see this relationship visualized in the following graph, where between the study hours of 7-7.5, the benefits of study begin to plateau:

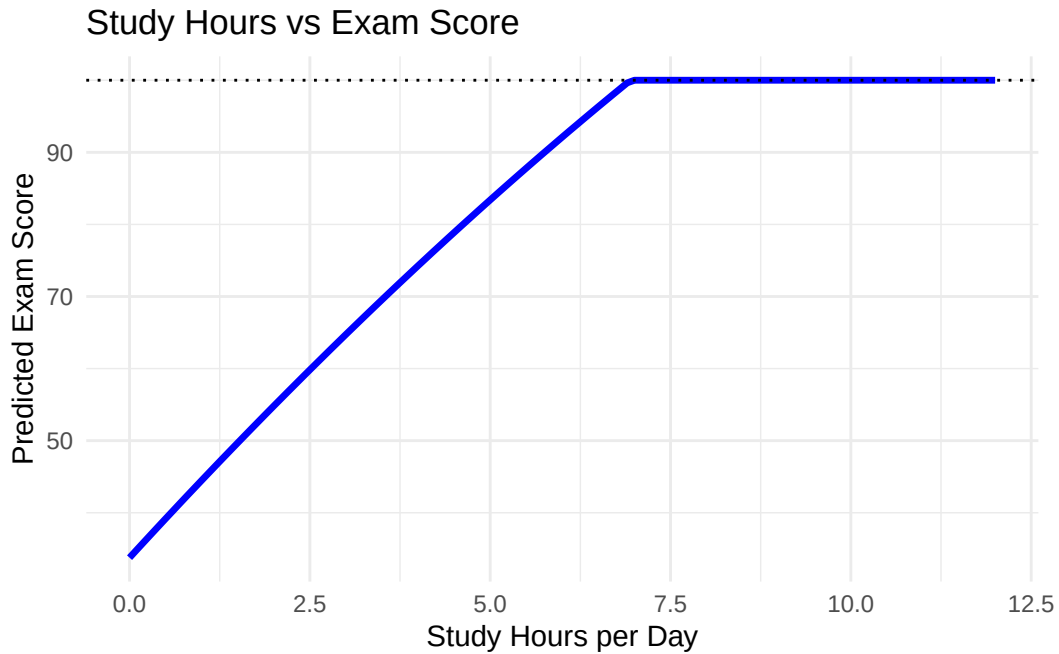


Figure 2: Predicted exam score as a function of study hours based on a quadratic regression model.

Once we knew that there were diminishing returns with study hours in relation to the predicted exam score, we wanted to evaluate if there was an interactive relationship between study hours and other statistically significant predictors for exam score. For our research, we focused on assessing whether the potential relationship between study hours and mental health was similar to the relationship between physical activity and mental health as described by Shaw et al. (2015). We evaluated two study hour models: a non-interactive quadratic model and an interactive quadratic model with mental health as the added predictor. After assessing our models with an ANOVA F-test, we found a statistically significant p-value of 0.01671

Table 4: Regression Results for Interaction Model

term	estimate	std.error	statistic	p.value
(Intercept)	24.106	2.516	9.581	< 2.2e-16
study_hours_per_day	9.366	1.429	6.552	9.086e-11
I(study_hours_per_day^2)	0.123	0.194	0.632	0.52779
mental_health_rating	1.752	0.406	4.311	1.786e-05
study_hours_per_day:mental_health_rating	0.297	0.229	1.296	0.19533
I(study_hours_per_day^2):mental_health_rating	-0.060	0.031	-1.935	0.05331

compared to our alpha of 0.05. This highlighted that the quadratic interaction between mental health and study hours is more significant to the resulting exam score than the non-interactive model. However, we would like to note an interesting discrepancy in our modeling results. Although the ANOVA comparison between the non-interactive and interactive models showed that there was a significant preference for the quadratic interaction, when we look at the p-value breakdown of its associated summary table, we see another story.

The output shows a statistically insignificant p-value between mental health and our quadratic study hours term, with a result of 0.0533. This result is very close to an alpha of 0.05, which currently makes our relational results between mental health and study hours inconclusive with the predicted exam score. From our model, we can therefore conclude that students who study more tend to score higher on exams, and the rate of improvement slightly increases as study time increases, but the gains begin to plateau. Linearly, students with better mental health also score higher on exams by 1.752 points. Additionally, on the interactive level, there's some evidence that better mental health may buffer against the potential negative effects of over-studying, but this effect is only borderline significant with the aforementioned p-value. So, while mental health is important for scoring well on exams, its interaction with study time needs more evidence to be conclusive.

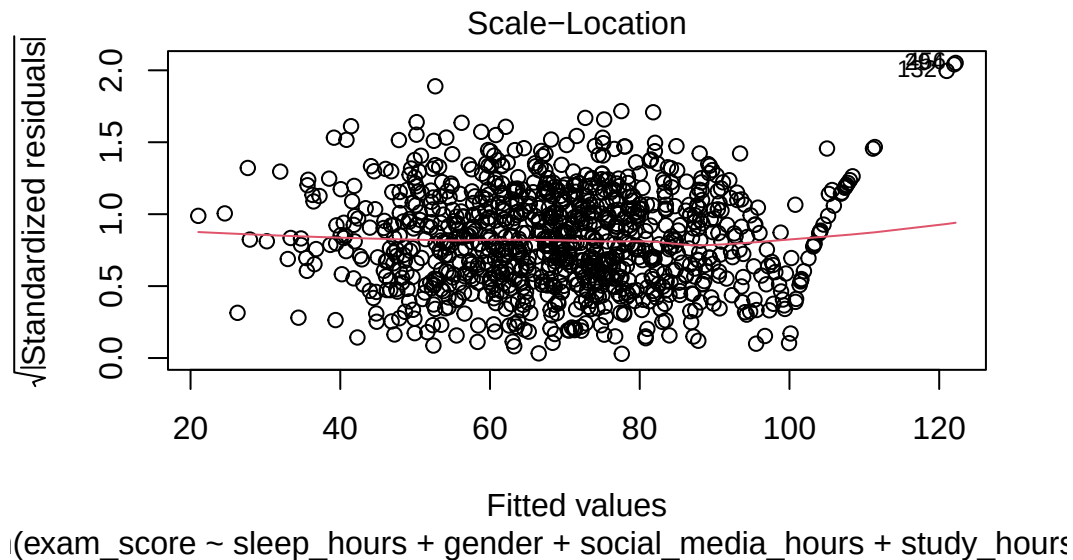
Discussion

Our study aimed to find what student behaviors were correlated with lower exam scores and if an interaction existed between these behaviors. To delve into this question, we honed in on the complicated relationship that could exist between time spent studying and a student's mental health. We conducted nested F-tests through ANOVA analyses and found that among our statistically significant predictors, the hours spent studying had a quadratic relationship with predicted exam score (see figure 2). This quadratic relation indicates that students should not over study, as there begins to be a plateau in exam score benefits. It was also found that there could be a quadratic relationship between the time spent studying and a student's mental health rating, which may influence predicted exam scores. This has important implications for a student's academic performance as the complicated relationships between a student's exam

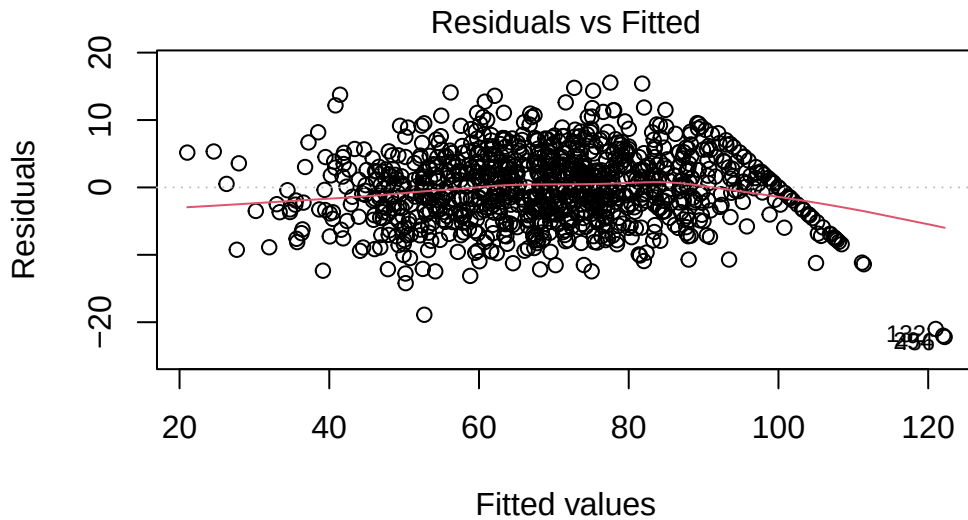
score, their daily hours of study time, and mental health could be relevant to the resources or assistance that students need to achieve their best possible academic performance. However, further research should be conducted to determine if this relationship is statistically significant, as our findings indicate that there is borderline significance of 0.0533 at the 0.05 alpha level. As described earlier in our paper, the data set we used is made up of simulated students and was not collected using real-life students. Therefore, replicating this study with a data set whose observations are real-life students is important to test if the findings and relationships we saw throughout our study still hold up. Perhaps using a data set composed of real-life students would have lessened some of the discrepancies that we saw throughout our analyses (particularly our interactive model) and could have had a more crucial impact on the attempts to fix the violations of our linearity conditions.

Data Analysis Appendix:

```
plot(full_model,3)
```



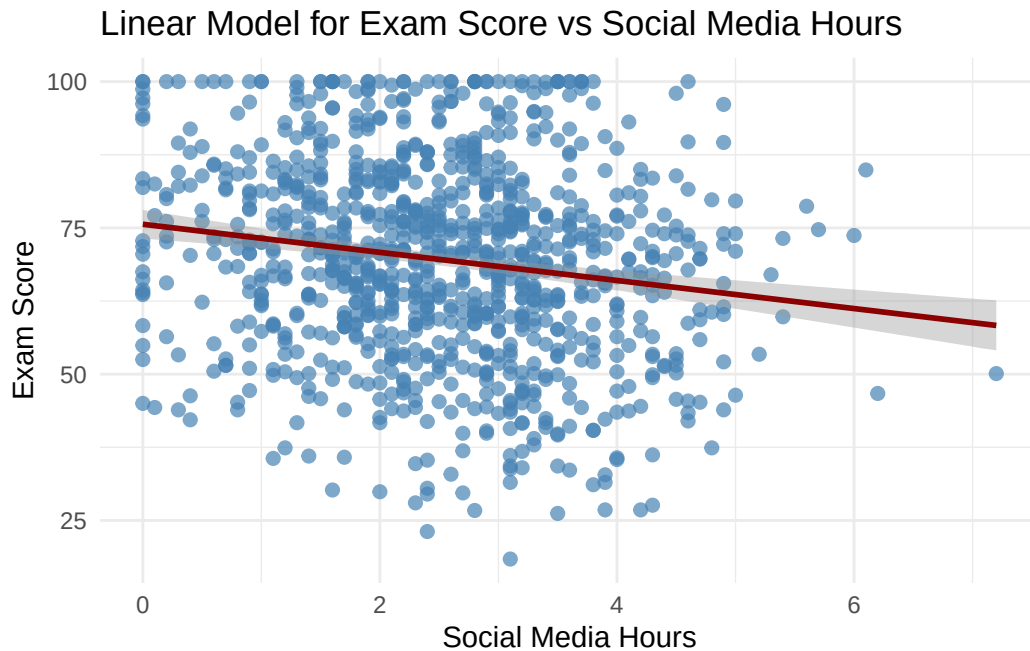
```
plot(full_model,1)
```



(exam_score ~ sleep_hours + gender + social_media_hours + study_hours)

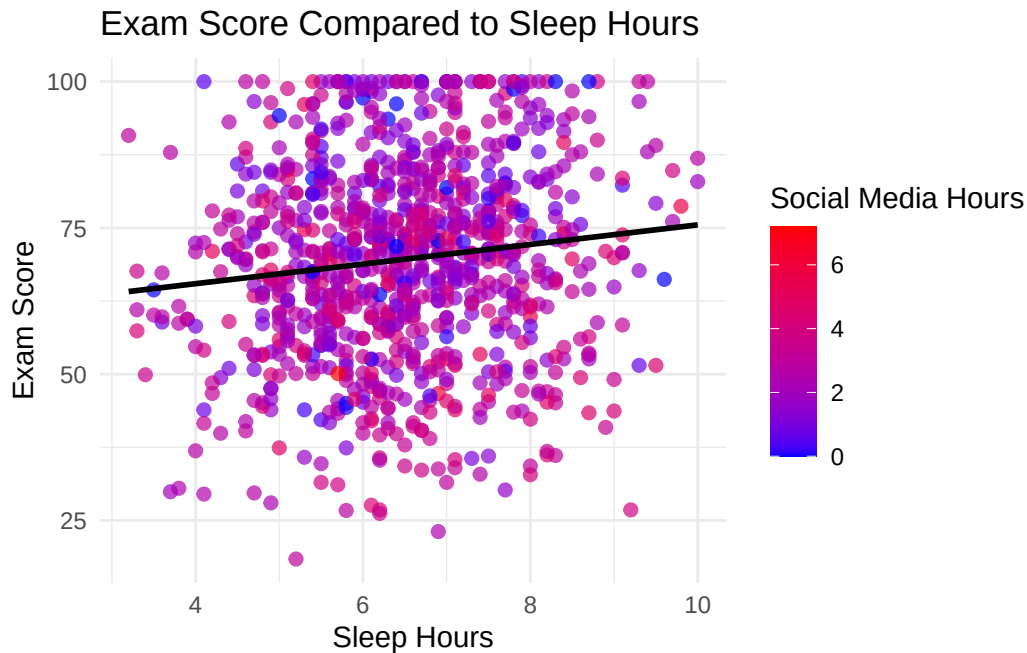
From both of these conditions for linearity and equal variance although they are not perfectly met we decided to continue with our original linear model with no transformation as log seemed to not improve our model.

```
ggplot(student_data, aes(x = social_media_hours, y = exam_score)) +
  geom_point(color = "steelblue", alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", se = TRUE, color = "darkred", linetype = "solid") +
  labs(
    title = "Linear Model for Exam Score vs Social Media Hours",
    x = "Social Media Hours",
    y = "Exam Score"
  ) +
  theme_minimal()
```



This graph shows the linear model relationship between Social Media Hours and Exam Score for exploration.

```
ggplot(student_data, aes(x = sleep_hours, y = exam_score, color = social_media_hours)) +  
  geom_point(size = 2, alpha = 0.7) +  
  scale_color_gradient(low = "blue", high = "red") +  
  geom_smooth(method = "lm", se = FALSE, color = "black") +  
  labs(  
    title = "Exam Score Compared to Sleep Hours",  
    x = "Sleep Hours",  
    y = "Exam Score",  
    color = "Social Media Hours"  
  ) +  
  theme_minimal()
```



In this data exploration each individual student in the data set is represented on the scatter plot. This graph showcases the relationship between their hours of sleep and hours of social media. From this relationship it then shows a resulting exam score. For example a student who got 4 hours of sleep and engaged with around 4 hours of social media received an exam score of almost 25. Meanwhile a student that received 4 hours of sleep but engaged in less than 2 hours of social media time achieved a score closer to 50.

Works Cited

- Centers for Disease Control and Prevention. 2019. "Dietary Behaviors and Academic Grades." <https://www.cdc.gov/healthy-schools/health-academics/diet-grades.html>.
- Nath, Jayanta. 2025. "Student Habits Vs Academic Performance." <https://www.kaggle.com/datasets/jayaantanaath/student-habits-vs-academic-performance>.
- Reuter, Peter R., and Bridget L. Forster. 2021. "Student Health Behavior and Academic Performance." *PeerJ* 9. <https://doi.org/10.7717/peerj.11107>.
- Shaw, Steven R., Paul Gomes, Anna Polotskaia, and Anna M. Jankowska. 2015. "The Relationship Between Student Health and Academic Performance: Implications for School Psychologists." *School Psychology International* 36 (2). <https://doi.org/10.1177/0143034314565425>.